CHAPTER EXERCISE: Answers for this chapter start on page 287.

A calculator should NOT be used on the following questions.

1

If $a + b = -2$, then $(a + b)^3 =$

A) 4
B) 0
C) -4
D) -8

2

For what value of n is $(n - 4)^2 = (n + 4)^2$?

3

If $\frac{1}{a} \times \frac{b}{c} = 1$, what is the value of $b - ac$?

A) -3
B) 0
C) 2
D) It cannot be determined from the information given.

4

If $3x - 8 = -23$, what is the value of $6x - 7$?

A) -5
B) -21
C) -30
D) -37

5

If $\frac{4}{9} = \frac{8}{3}m$, what is the value of m ?

A) $\frac{1}{6}$
B) $\frac{2}{3}$
C) $\frac{5}{6}$
D) 6

6

If $3x + 1 = -8$, what is the value of $(x + 2)^3$?

A) -1
B) 1
C) 8
D) 125

7

If $\frac{4}{k+2} = \frac{x}{3}$, where $k \neq -2$, what is k in terms of x ?

A) $\frac{12 - 2x}{x}$
B) $\frac{12 + 2x}{x}$
C) $\frac{x}{12 + 2x}$
D) $12x - 2$

8

If $(x - 3)^2 = 36$ and $x < 0$, what is the value of x^2 ?

9

$$f = p\left(\frac{(1+i)^n - 1}{i}\right)$$

The formula above gives the future value f of an annuity based on the monthly payment p, the interest rate i, and the number of months n. Which of the following gives p in terms of f, i, and n ?

A) $\dfrac{fi}{(1+i)^n - 1}$

B) $\dfrac{(1+i)^n - 1}{fi}$

C) $\dfrac{f - i}{(1+i)^n - 1}$

D) $fi + 1 - (1+i)^n$

10

If $\dfrac{m}{2n} = 2$, what is the value of $\dfrac{n}{2m}$?

A) $\dfrac{1}{8}$

B) $\dfrac{1}{4}$

C) $\dfrac{1}{2}$

D) 1

11

$$x^2 + 5x - 24 = 0$$

If k is a solution of the equation above and $k < 0$, what is the value of $|k|$?

12

If $y > 0$ and $\left(\dfrac{y}{2}\right)^3 = \dfrac{y}{32}$, what is the value of y ?

13

If $\dfrac{2\sqrt{x+4}}{3} = 6$ and $x > 0$, what is the value of x ?

14

$$20 - \sqrt{x} = \frac{2}{3}\sqrt{x} + 10$$

If $x > 0$, for what value of x is the equation above true?

15

If $x + y = \sqrt{x^2 + y^2 + 16}$, what is the value of xy ?

16

$$\frac{2x-1}{x+2} = \frac{x-2}{4}$$

What is the solution set to the equation above?

A) $\{-10, 0\}$
B) $\{-10, -4\}$
C) $\{0, 8\}$
D) $\{-4, 8\}$

17

If $x > 0$ and $\left(\frac{x}{6}\right)^2 - 2\left(\frac{x}{6}\right) - 15 = 0$, what is the value of x ?

18

$$\frac{x^2 - 4x + 3}{x - 1} = 4$$

What is the solution to the equation above?

19

$$x^2(x^4 - 9) = 8x^4$$

If $x > 0$, for what real value of x is the equation above true?

A calculator is allowed on the following questions.

20

$$y + 2kx = kx^2 + 5$$

In the equation above, k is a constant. If $y = 23$ when $x = 3$, what is the value of k ?

A) -6
B) 3
C) 6
D) 9

21

If $\frac{x}{6} = \frac{x+12}{42}$, what is the value of $\frac{6}{x}$?

A) $\frac{1}{3}$
B) 2
C) 3
D) 6

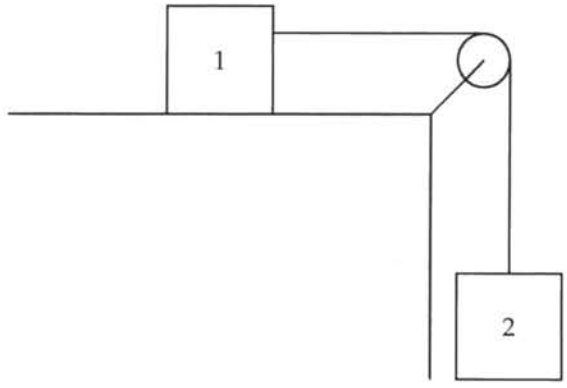
22

$$d = a\left(\frac{c+1}{24}\right)$$

Doctors use Cowling's rule, shown above, to determine the right dosage d, in milligrams, of medication for a child based on the adult dosage a, in milligrams, and the child's age c, in years. Ben is a patient who is in need of a certain medication. If a doctor uses Cowling's rule to prescribe Ben a dosage that is half the adult dosage, what is Ben's age, in years?

A) 7
B) 9
C) 11
D) 13

Questions 23-24 refer to the following information.



In the figure above, two objects are connected by a string which is threaded through a pulley. Using its weight, object 2 moves object 1 along a flat surface. The acceleration a of the two objects can be determined by the following formula

$$a = \frac{m_2 g - \mu m_1 g}{m_1 + m_2}$$

where m_1 and m_2 are the masses of object 1 and object 2, respectively, in kilograms, g is the acceleration due to Earth's gravity measured in $\frac{\text{m}}{\text{sec}^2}$, and μ is a constant known as the coefficient of friction.

23

Which of the following expresses μ in terms of the other variables?

A) $\mu = \dfrac{a(m_1 + m_2)}{m_1 m_2 g^2}$

B) $\mu = \dfrac{a(m_1 + m_2)}{m_2 g - m_1 g}$

C) $\mu = \dfrac{m_2 g - a(m_1 + m_2)}{m_1 g}$

D) $\mu = \dfrac{a(m_1 + m_2) - m_2 g}{m_1 g}$

24

If the masses of both object 1 and object 2 were doubled, how would the acceleration of the two objects be affected?

A) The acceleration would stay the same.

B) The acceleration would be halved.

C) The acceleration would be doubled.

D) The acceleration would be quadrupled (multipled by a factor of 4).

25

If $3(x - 2y) - 3z = 0$, which of the following expresses x in terms of y and z ?

A) $\dfrac{2y + 3z}{3}$

B) $2y + z$

C) $y + 2z$

D) $6y + 3z$

26

$$(x + 1)(x - 2) = 7x - 18$$

If x is the solution to the equation above, what is the value of $7x - 18$?

27

$$2\sqrt{x} = x - 3$$

Which of the following represents all the possible values of x that satisfy the equation above?

A) 1 and 9

B) 1 and 4

C) 4

D) 9

28

$$\frac{4}{x^2 - 6x + 9} = 9$$

Based on the equation above, which of the following could be the value of $x - 3$?

A) $\frac{2}{3}$

B) $\frac{3}{2}$

C) $\frac{7}{3}$

D) $\frac{9}{2}$

29

$$\sqrt{x - 10} = \sqrt{x} - \sqrt{2}$$

In the equation above, what is the value of $\sqrt{x - 10}$?

A) $\sqrt{6}$

B) $2\sqrt{2}$

C) $3\sqrt{2}$

D) $\sqrt{14}$

30

$$xy^2 + x - y^2 - 1 = 0$$

If the equation above is true for all real values of y, what must the value of x be?

Questions 31-32 refer to the following information.

$$V = P(1 - r)^t$$

The value V of a car depreciates over t years according to the formula above, where P is the original price and r is the annual rate of depreciation.

31

Which of the following expresses r in terms of V, P, and t?

A) $r = 1 - \sqrt[t]{\frac{V}{P}}$

B) $r = 1 + \sqrt[t]{\frac{V}{P}}$

C) $r = \sqrt[t]{\frac{V}{P}} - 1$

D) $r = 1 - \frac{\sqrt[t]{V}}{P}$

32

If a car depreciates to a value equal to half its original price after 5 years, then which of the following is closest to the car's annual rate of depreciation?

A) 0.13

B) 0.15

C) 0.16

D) 0.2

CHAPTER EXERCISE:

1. $\boxed{D}$

$$(a+b)^3 = (-2)^3 = -8$$

2. $\boxed{0}$ The answer should be obvious just by looking at it. Testing $n = 0$ gives us:

$$(0-4)^2 = (0+4)^2$$
$$(-4)^2 = (4)^2$$
$$16 = 16$$

We could also expand and solve like so:

$$(n-4)(n-4) = (n+4)(n+4)$$
$$n^2 - 8n + 16 = n^2 + 8n + 16$$
$$-16n = 0$$
$$n = 0$$

3. $\boxed{B}$

$$\frac{b}{ac} = 1$$
$$b = ac$$

If $b = ac$, then $b - ac$ must equal 0.

4. $\boxed{D}$ If $3x - 8 = -23$, then $3x = -15$. Multiplying both sides by 2, $6x = -30$ and $6x - 7 = -37$.

5. $\boxed{A}$ Cross multiply.

$$\frac{4}{9} = \frac{8}{3}m$$
$$12 = 72m$$
$$\frac{1}{6} = m$$

6. $\boxed{A}$

$$3x + 1 = -8$$
$$3x = -9$$
$$x = -3$$

$$(x+2)^3 = (-3+2)^3 = (-1)^3 = -1$$

7. $\boxed{A}$ Cross multiply to get $12 = kx + 2x$. Then, $k = \dfrac{12 - 2x}{x}$

8. $\boxed{9}$ Note that

$$(-6)^2 = 36$$

which happens when $x = -3$. Then $x^2 = (-3)^2 = 9$.

9. $\boxed{A}$

$$f = p\left(\frac{(1+i)^n - 1}{i}\right)$$
$$fi = p((1+i)^n - 1)$$
$$\frac{fi}{(1+i)^n - 1} = p$$

10. $\boxed{A}$ Multiply both sides by 2 to get $\dfrac{m}{n} = 4$, which means $\dfrac{n}{m} = \dfrac{1}{4}$. Then,

$$\frac{n}{2m} = \frac{1}{2} \cdot \frac{n}{m} = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$$

11. $\boxed{8}$ $x^2 + 5x - 24 = 0$ can be factored as $(x+8)(x-3) = 0$. The two possible solutions are then -8 and 3. Since $k < 0$, $k = -8$ and $|k| = 8$.

12. $\boxed{\dfrac{1}{2}}$

$$\left(\frac{y}{2}\right)^3 = \frac{y}{32}$$
$$\frac{y^3}{8} = \frac{y}{32}$$
$$y^2 = \frac{8}{32}$$
$$y = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

13. $\boxed{77}$

$$\frac{2\sqrt{x+4}}{3} = 6$$
$$2\sqrt{x+4} = 18$$
$$\sqrt{x+4} = 9$$
$$x+4 = 81$$
$$x = 77$$

14. $\boxed{36}$

$$20 - \sqrt{x} = \frac{2}{3}\sqrt{x} + 10$$
$$10 = \frac{5}{3}\sqrt{x}$$
$$6 = \sqrt{x}$$
$$36 = x$$

15. $\boxed{8}$ Square both sides of the equation.

$$(x+y)^2 = \left(\sqrt{x^2+y^2+16}\right)^2$$
$$x^2 + 2xy + y^2 = x^2 + y^2 + 16$$
$$2xy = 16$$
$$xy = 8$$

16. $\boxed{C}$ Cross multiply and expand both sides.

$$4(2x-1) = (x+2)(x-2)$$
$$8x - 4 = x^2 - 4$$
$$0 = x^2 - 8x$$
$$0 = x(x-8)$$
$$x = 0, 8$$

Neither are false solutions, so the solution set is $\{0, 8\}$.

17. $\boxed{30}$ To make this problem easier to work with, let $A = \frac{x}{6}$. Then,

$$\left(\frac{x}{6}\right)^2 - 2\left(\frac{x}{6}\right) - 15 = 0$$
$$A^2 - 2A - 15 = 0$$
$$(A-5)(A+3) = 0$$
$$A = 5, -3$$

So $\frac{x}{6} = 5$ or $\frac{x}{6} = -3$. Solving these equations gives $x = 30$ and $x = -18$. Since the question specifies that $x > 0$, the answer is 30.

18. $\boxed{7}$ There are two ways to approach this problem. The faster way is to factor the numerator first.

$$\frac{x^2 - 4x + 3}{x - 1} = 4$$
$$\frac{(x-3)\cancel{(x-1)}}{\cancel{x-1}} = 4$$
$$x - 3 = 4$$
$$x = 7$$

The second way is to get rid of the fraction first by multiplying both sides by $x - 1$ and then factor later.

$$\frac{x^2 - 4x + 3}{x - 1} = 4$$
$$x^2 - 4x + 3 = 4(x-1)$$
$$x^2 - 4x + 3 = 4x - 4$$
$$x^2 - 8x + 7 = 0$$
$$(x-7)(x-1) = 0$$
$$x = 7, 1$$

Now, $x = 1$ is a false solution since it causes division by 0. Therefore, the solution is $x = 7$.

19. [3] Because this is a no calculator question, guess and check is a valid strategy. You can also do the following:

$$x^2(x^4-9) = 8x^4$$
$$x^6 - 9x^2 - 8x^4 = 0$$
$$x^2(x^4 - 8x^2 - 9) = 0$$
$$x^2(x^2-9)(x^2+1) = 0$$
$$x^2(x+3)(x-3)(x^2+1) = 0$$

Because $x > 0$, x must be 3 for the equation above to be true.

20. [C] Plug in the given x and y values.

$$y + 2kx = kx^2 + 5$$
$$23 + 2(k)(3) = k(3)^2 + 5$$
$$23 + 6k = 9k + 5$$
$$-3k = -18$$
$$k = 6$$

21. [C] Cross multiply.

$$\frac{x}{6} = \frac{x+12}{42}$$
$$42x = 6x + 72$$
$$36x = 72$$
$$x = 2$$

Now, $\frac{6}{x} = \frac{6}{2} = 3$.

22. [C]

$$d = a\left(\frac{c+1}{24}\right)$$
$$\frac{a}{2} = a\left(\frac{c+1}{24}\right)$$
$$\frac{1}{2} = \frac{c+1}{24}$$
$$12 = c+1$$
$$11 = c$$

23. [C]

$$a = \frac{m_2 g - \mu m_1 g}{m_1 + m_2}$$
$$a(m_1+m_2) = m_2 g - \mu m_1 g$$
$$a(m_1+m_2) - m_2 g = -\mu m_1 g$$
$$\frac{m_2 g - a(m_1+m_2)}{m_1 g} = \mu$$

24. [A] Because the 2's cancel out, the acceleration stays the same.

$$a_{new} = \frac{2m_2 g - \mu(2m_1)g}{2m_1 + 2m_2} = \frac{2(m_2 g - \mu m_1 g)}{2(m_1+m_2)}$$
$$= a_{old}$$

25. [B]

$$3(x-2y) - 3z = 0$$
$$3x - 6y - 3z = 0$$
$$3x = 6y + 3z$$
$$x = 2y + z$$

26. [10] First, expand the left hand side. Then combine like terms and factor.

$$(x+1)(x-2) = 7x - 18$$
$$x^2 - x - 2 = 7x - 18$$
$$x^2 - 8x + 16 = 0$$
$$(x-4)^2 = 0$$

Therefore, $x = 4$ and $7x - 18 = 7(4) - 18 = 10$.

27. [D] We can either plug in the answer choices or solve algebraically. Plugging in the answer choices is more efficient here, but since that's self-explanatory, let's solve algebraically. First, square both sides.

$$\begin{aligned}(2\sqrt{x})^2 &= (x-3)^2\\ 4x &= x^2-6x+9\\ 0 &= x^2-10x+9\\ 0 &= (x-1)(x-9)\\ x &= 1,9\end{aligned}$$

Since 1 is a false solution, the only value of x that satisfies the equation is 9.

28. [A]

$$\begin{aligned}\frac{4}{x^2-6x+9} &= 9\\ \frac{4}{(x-3)^2} &= 9\\ \frac{4}{9} &= (x-3)^2\\ \pm\sqrt{\frac{4}{9}} &= \sqrt{(x-3)^2}\\ \pm\frac{2}{3} &= x-3\end{aligned}$$

29. [B] We can use the answer choices to backsolve or we can solve the equation algebraically. We'll first solve it algebraically by squaring both sides.

$$\begin{aligned}(\sqrt{x-10})^2 &= (\sqrt{x}-\sqrt{2})^2\\ x-10 &= (\sqrt{x})^2-2(\sqrt{x})(\sqrt{2})+(\sqrt{2})^2\\ x-10 &= x-2\sqrt{2x}+2\\ -12 &= -2\sqrt{2x}\\ 6 &= \sqrt{2x}\\ 36 &= 2x\\ 18 &= x\end{aligned}$$

Let's say we wanted to test the answer choices instead. How would we do that? For choice A, if $\sqrt{6}$ is the value of $\sqrt{x-10}$, then x would have to equal 16. If we plug $x=16$ into the equation, we get $4=4-\sqrt{2}$, which is false. We can eliminate A. For choice B, $2\sqrt{2}=\sqrt{8}$, so if that's the value of $\sqrt{x-10}$, then x would have to equal 18. If we plug $x=18$ into the equation, the left hand side is $\sqrt{18-10}=2\sqrt{2}$ and the right hand side is $\sqrt{18}-\sqrt{2}=3\sqrt{2}-\sqrt{2}=2\sqrt{2}$. Since both sides match, choice B must be the answer.

30. [1]

$$\begin{aligned}xy^2+x-y^2-1 &= 0\\ x(y^2+1)-(y^2+1) &= 0\\ (y^2+1)(x-1) &= 0\end{aligned}$$

Since y^2+1 is always positive, x must equal 1.

31. [A] Divide both sides by P and take the tth root of both sides.

$$\begin{aligned}V &= P(1-r)^t\\ \frac{V}{P} &= (1-r)^t\\ \sqrt[t]{\frac{V}{P}} &= 1-r\\ r &= 1-\sqrt[t]{\frac{V}{P}}\end{aligned}$$

32. [A] From the previous question, we know that $r=1-\sqrt[t]{\frac{V}{P}}$. Because V is half P, $\frac{V}{P}=\frac{1}{2}$. Thus, $r=1-\sqrt[5]{\frac{1}{2}}\approx 0.13$